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St. Joseph's College of Arts & Science (Autonomous)

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B.SC DEGREE EXAMINATION – APRIL 2021

EMB615S - COMPUTERS APPLICATIONS IN BIOLOGY

Time : 3 hrs **Max Marks :75**

SECTION – A (10X1=10)

Answer **ALL** the Questions

Choose the correct answer

1. Computer or system peripherals that receives data from processing unit are called
   a) Output devices b) Receiver
   c) Preprinted documents d) analogue devices
2. A CPU contains
   a) A card reader and a printing device
   b) an analytical engine and a control unit
   c) A control unit and an arithmetic logic unit
   d) An arithmetic logic unit and a card unit
3. Which of the following is the first biological data base?
   a) Gen Bank b) DDBJ
   c) Atlas of protein sequence and structure d) OMIM
4. Which of the following is the search and retrieval tool of NCBI?
   a) SAKURA b) SRS c) Sequin d) Entréz
5. Alignment method suitable for aligning closely related sequence is
   a) Multiple b) Pair wise alignment
   c) Global alignment d) local alignment

6. Sequence alignment helps scientist

   a) To trace out evolutionary relationships
   b) To infer the functions of newly synthesized genes
   c) To predict new members of gene families
   d) All the above

7. Human genome project was also focused on identifying

   a) SNPS b) VNTRs c) Mini satellites d) Junk DNA

8. How are the four different bases distinguished in automated sequencing systems?

   a) Each base has a different radioactive tag
   b) Each has a distinctive fluorescent tag
   c) Each base has a unique antibody bound to it
   d) All the above

9. Which types of genomics studies the physical nature of genomes?

   a) Comparative genomics b) structural genomics
   c) functional genomics d) all the above

10. The DNA microarrays technology that tracks deletions and amplifications of specific DNA sequence is called _____

    a) DNA variation screening
    b) Gene expresson profiling
    c) Microarray comparative genomic hybridization
    d) d) antisense

SECTION – B (5X3=15)

Answer any **FIVE** Questions

11. What is super computer?
12. Write a note on structure data base, PDB

13. What are the advantages of Clustal W tool?
14. Write a note on the tools of phylogenetic analysis
15. What are the areas of proteomics?
16. What do you understand by single nucleotide polymorphism (SNP)
17. Give an account on the types of proteomics

SECTION – C (5X10=50)

Answer **ALL** the Questions

18. a) Write an account on the uses of internet.

(Or)

b) Give an account on the generations of computers

19. a) Give a detailed account on the history of bio informatics.

(Or)

b) Discuss in detail on the different data bases of nucleic acid sequence

20. a) Differentiate global and local alignment.

(Or)

b) Give an account on the BLAST variants of different searches

21. a) Describe the various molecular modeling visualizing tools with special reference to RasMol.

(Or)

b) Give a detailed account on automated DNA sequencing methods

22. a) Describe the technique of DNA micro array with illustrations.

(Or)

b) Define genomics. List out the types of genomics and explain structural genomics

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